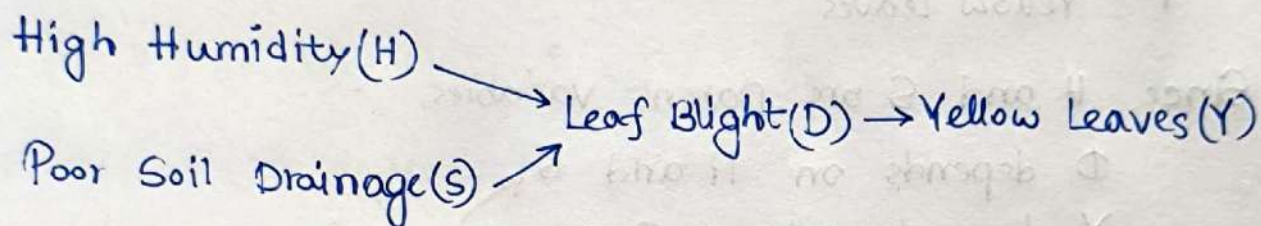


- ① A farmer is monitoring a tomato field for a fungal disease called Leaf Blight (D). Two factors influence whether the disease occurs:

High Humidity (H) and Poor Soil Drainage (S)

If the disease is present, the plant is more likely to show Yellow Leaves (Y).

The Bayesian Network is shown below:



Variable	State	Probability
H	True	0.6
	False	0.4
S	True	0.3
	False	0.7

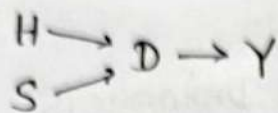
H	S	$P(D=True)$
True	True	.90
True	False	.70
False	True	.60
False	False	.10

D	$P(Y=True)$
True	0.85
False	0.20

- ② Find the joint probability that: High Humidity is present, Poor Soil Drainage is present, Leaf Blight occurs, and Yellow Leaves are observed.

Answer:

Given Bayes' Network:



Where,

H = High Humidity

S = Poor Soil Drainage

D = Leaf Blight

Y = Yellow Leaves

Since H and S are parent variables,

D depends on H and S

Y depends only on D

Therefore,

$$P(H, S, D, Y) = P(H) \cdot P(S) \cdot P(D | H, S) \cdot P(Y | D)$$

Given probabilities for Parent Variables:

High Humidity (H)

$$P(H) = 0.6$$

$$P(\neg H) = 0.4$$

Poor Soil Drainage (S)

$$P(S) = 0.3$$

$$P(\neg S) = 0.7$$

Given probability of Leaf Blight:

H	S	$P(D H, S)$
True	True	0.9
True	False	0.7
False	True	0.6
False	False	0.1

D	$P(Y = \text{True})$
True	0.85
False	0.20

$$\begin{aligned} \therefore P(H, S, D, Y) &= 0.6 \times 0.3 \times 0.9 \times 0.85 \\ &= 0.1377 \end{aligned}$$

⑥ Find the probability that yellow leaves are observed.

Answer:

We need, $P(Y = \text{True})$

Since Y depends on D , we need to consider both possibilities:

- ① Leaf blight occurs
- ② Leaf blight does not occur

Using the law of total probability:

$$P(Y) = P(Y|D) \cdot P(D) + P(Y|\neg D) \cdot P(\neg D)$$

However, $P(D)$ depends on H and S .

Now there are four possible combinations of H and S .

Case 1: $H = \text{True}, S = \text{True}$

$$\begin{aligned} P(H, S, D) &= P(H) \cdot P(S) \cdot P(D|H, S) \\ &= 0.6 \times 0.3 \times 0.9 \\ &= 0.162 \end{aligned}$$

Case 2: $H = \text{True}, S = \text{False}$

$$\begin{aligned} P(H, \neg S, D) &= P(H) \times P(\neg S) \times P(D|H, \neg S) \\ &= 0.6 \times 0.7 \times 0.7 \\ &= 0.294 \end{aligned}$$

Case 3: $H = \text{False}, S = \text{True}$

$$\begin{aligned} P(\neg H, S, D) &= P(\neg H) \times P(S) \times P(D|\neg H, S) \\ &= 0.4 \times 0.3 \times 0.6 \\ &= 0.07 \end{aligned}$$

Case 4: $H = \text{False}, S = \text{False}$

$$\begin{aligned} P(\neg H, \neg S, D) &= P(\neg H) \times P(\neg S) \times P(D|\neg H, \neg S) \\ &= 0.4 \times 0.7 \times 0.1 \\ &= 0.028 \end{aligned}$$

Adding all four cases to compute $P(D)$,

$$P(D) = 0.162 + 0.294 + 0.072 + 0.028 \\ = 0.556$$

$$\therefore P(\neg D) = 1 - 0.556 \\ = 0.444$$

We know,

D	$P(Y=\text{True})$
True	0.85
False	0.20

Therefore,

$$P(Y=\text{True}) = P(Y|D) \cdot P(D) + P(Y|\neg D) \cdot P(\neg D) \\ = 0.85 \times 0.556 + 0.2 \times 0.444 \\ = 0.5614$$

②

Sample	Weather	Temp	Windy	Play Outside
1	Sunny	Hot	No	Yes
2	Sunny	Hot	Yes	No
3	Cloudy	Mild	No	Yes
4	Rainy	Cool	No	Yes
5	Rainy	Cool	Yes	No
6	Cloudy	Cool	Yes	Yes
7	Sunny	Mild	No	Yes
8	Rainy	Mild	No	Yes

① Calculate prior probabilities of the classes:

Play Outside = Yes

Play Outside = No

Answer:

We know,

The prior probability of a class is:

$$P(\text{class}) = \frac{\text{Number of Samples belonging to class}}{\text{Total Number of Samples}}$$

Total number of samples = 8

'Yes' samples = 6

$$\therefore P(\text{Play Outside} = \text{Yes}) = \frac{6}{8} = 0.75$$

'No' samples = 2

$$\therefore P(\text{Play Outside} = \text{No}) = \frac{2}{8} = 0.25$$

⑥ Using Naive Bayes Classifier, compute the posterior probabilities:

i) $P(\text{Yes} \mid \text{Rainy, Cool, No})$

ii) $P(\text{No} \mid \text{Rainy, Cool, No})$

Answer:

The query, $X = (\text{Weather} = \text{Rainy},$
 $\text{Temperature} = \text{Cool},$
 $\text{Windy} = \text{No})$

For this query, the Naive Bayes Formula becomes:

$$P(C \mid X) \propto P(C) P(X_1 \mid C) P(X_2 \mid C) P(X_3 \mid C)$$

Where,

$C = \text{Class (Yes/No)}$

$X_1 = \text{Rainy}$

$X_2 = \text{Cool}$

$X_3 = \text{No}$

i) $P(\text{Yes} \mid \text{Rainy, Cool, No})$

From 'a': $P(\text{Yes}) = 0.75$

$$P(X_1 \mid \text{Yes})$$

$$\Rightarrow P(\text{Rainy} \mid \text{Play Outside} = \text{Yes})$$

Here, we only look at the rows where play outside = Yes
There are 6 'Yes' rows.

Among them, weather = Rainy appears two times.

$$\therefore P(\text{Rainy} \mid \text{Yes}) = \frac{2}{6}$$

$$\text{Similarly, } P(X_2 \mid \text{Yes}) = P(\text{Cool} \mid \text{Yes}) = \frac{2}{6}$$

$$P(X_3 \mid \text{Yes}) = P(\text{No} \mid \text{Yes}) = \frac{5}{6}$$

$$\therefore P(\text{Yes} | \text{Rainy}, \text{Cool}, \text{No}) \propto$$

$$P(\text{Yes}) \cdot P(\text{Rainy} | \text{Yes}) \cdot P(\text{Cool} | \text{Yes}) \cdot P(\text{No} | \text{Yes})$$

$$\begin{aligned} \Rightarrow P(\text{Yes} | \text{Rainy}, \text{Cool}, \text{No}) &= 0.75 \times \frac{2}{6} \times \frac{2}{6} \times \frac{5}{6} \\ &= \frac{5}{72} \\ &\approx 0.0694 \end{aligned}$$

$$\textcircled{\text{ii}} P(\text{No} | \text{Rainy}, \text{Cool}, \text{No})$$

$$\text{From 'a': } P(\text{No}) = 0.25$$

$$P(\text{Not Rainy})$$

$$P(\text{Rainy} | \text{No}) = \frac{1}{2}$$

$$P(\text{Cool} | \text{No}) = \frac{1}{2}$$

$$P(\text{No} | \text{No}) = \frac{0}{2}$$

$$\therefore P(\text{No} | \text{Rainy}, \text{Cool}, \text{No}) \propto$$

$$P(\text{No}) \cdot P(\text{Rainy} | \text{No}) \cdot P(\text{Cool} | \text{No}) \cdot P(\text{No} | \text{No})$$

$$\begin{aligned} \Rightarrow P(\text{No} | \text{Rainy}, \text{Cool}, \text{No}) &= 0.25 \times \frac{1}{2} \times \frac{1}{2} \times 0 \\ &= 0 \end{aligned}$$

Normalization:

$$P(\text{Yes} | X) = \frac{P(\text{Yes} | X)}{P(\text{Yes} | X) + P(\text{No} | X)} = \frac{\frac{5}{72}}{\frac{5}{72} + 0} = 1$$

$$P(\text{No} | X) = \frac{P(\text{No} | X)}{P(\text{Yes} | X) + P(\text{No} | X)} = \frac{0}{\frac{5}{72} + 0} = 0$$

© Based on your calculations, predict whether the children will play outside.

Answer:

From 2(b), Posterior probabilities:

$$P(\text{Yes} | \text{Rainy}, \text{Cool}, \text{No}) = 1$$

$$P(\text{No} | \text{Rainy}, \text{Cool}, \text{No}) = 0$$

Since $1 > 0$,

the Naive Bayes Classifier predicts

Play outside = Yes

The normalized posterior probability of 100%, suggests that the children will play outside.